Swin Transformer: Hierarchical Vision Transformer using Shifted Windows

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Abstract

This paper investigates architectures with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction

Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding Swin Transformer: Hierarchical Vision Transformer using Shifted Windows.

Section topic: Architectures
Question context: How does Swin Transformer introduce hierarchical representations into vision backbones?
Key Finding: Swin Transformer limits self-attention to local windows and shifts window boundaries between consecutive layers to enable cross-window connections with linear image-size scaling [Swin Transformer, p. 3].
Evidence passage: Computes self-attention within non-overlapping local windows Shifted window partitioning allows cross-window connections with linear computational complexity

2. Technical Methodology & Empirical Findings

Section topic: Multi-Paper Analysis
Question context: How do Swin Transformer and original ViT differ in computational scaling relative to image resolution?
Key Finding: ViT scales quadratically with image resolution [An Image is Worth 16x16 Words, p. 3], while Swin Transformer achieves linear complexity by computing attention within shifted local windows [Swin Transformer, p. 2].
Evidence passage: ViT has quadratic computational complexity relative to image resolution Swin Transformer has linear computational complexity due to local window self-attention

3. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

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